

Log-Log Scaling Law of Terminal Metrics vs ϵ_g

Osc slope ≈ 0.67 , d_{slow} slope ≈ 0.98 (first-order=1, gap from $\sigma(\Delta)$ saturation)

Terminal Metrics

Monotone Scaling Preserved:

$\epsilon_g \uparrow 10^2 \Rightarrow d_{\text{slow}} \uparrow 91.0x$

Sub-first-order reflects $\sigma(\Delta)$ softmax higher-order residual

$\blacksquare$ Osc($x; T_w$), slope=0.67
 $\bullet$ $d_{\text{slow}}(T)$, slope=0.98
 $- -$ $O(\epsilon_g)$ Reference (slope=1, strict only as $\epsilon_g \rightarrow 0$)

